

FILTER DESIGN TEST CASE

Lattice Filter - Band 66 (AWS) for IoT Narrowband Module
Design Validation and Performance Characterization

DOCUMENT INFORMATION

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1. OBJECTIVE

This document describes the design, simulation, and characterization of a Lattice Filter filter targeting Band 66 (AWS) for IoT Narrowband Module. The filter is designed on PZT on Si substrate with a target center frequency of 3986.7 MHz and bandwidth of 323.8 MHz. The objective is to validate that the filter meets the required insertion loss, rejection, and linearity specifications for integration into the IoT Narrowband Module module. This design iteration addresses feedback from the previous revision regarding out-of-band rejection improvements and temperature stability.

2. SCOPE

- Filter Type: Lattice Filter
- Frequency Band: Band 66 (AWS)
- Application: IoT Narrowband Module
- Substrate: PZT on Si
- Package: QFN (Quad Flat No-Lead)
- Design Tool: CST Studio Suite 2024
- Design Center: San Jose Advanced Design Lab

This specification applies to all variants of the Lattice Filter design targeting Band 66 (AWS). Results are used to qualify the design for mass production release.

3. DESIGN PARAMETERS

Center Frequency:	3986.7 MHz
Bandwidth (3 dB):	323.8 MHz
Insertion Loss Target:	≤ 2.8 dB
Return Loss Target:	≥ 20.0 dB
Out-of-Band Rejection:	≥ 28.0 dB
Isolation (Duplexer):	≥ 27.0 dB
Q Factor:	1148
Temperature Coefficient:	-14.4 ppm/°C
Die Size:	2.6 x 0.7 mm
Wafer Size:	8-inch
Number of Resonators:	11
Electrode Material:	Mo

Passivation: SiO2/Si3N4 stack

4. TEST PROCEDURE

1. Open CST Studio Suite 2024 and load the Lattice Filter template project.
2. Define the substrate stack: PZT on Si with specified layer thicknesses.
3. Set design targets: center freq 3986.7 MHz, BW 323.8 MHz.
4. Run initial EM simulation with 2D mesh density of 20 cells/wavelength.
5. Optimize resonator geometry using gradient descent (target IL < 2.8 dB).
6. Verify coupling coefficients between resonator stages.
7. Run 3D FEM simulation to account for package parasitics (QFN (Quad Flat No-Lead)).
8. Export GDS-II layout for mask fabrication.
9. Fabricate prototype wafers (8-inch, PZT on Si).
10. Perform on-wafer probing using Rohde & Schwarz ZNA67.
11. Measure S-parameters (S11, S21, S12, S22) from 100 MHz to 11960 MHz.
12. Compare measured results against simulation and specification limits.
13. Perform temperature cycling (-40°C to +85°C) and re-measure.
14. Document results and generate summary report.

5. ACCEPTANCE CRITERIA

- Insertion loss at center frequency shall be <= 2.8 dB
- Return loss in passband shall be >= 20.0 dB
- Out-of-band rejection at +/- 486 MHz offset >= 28.0 dB
- Group delay variation across passband shall be <= 3 ns
- Temperature drift over -40°C to +85°C shall be <= 7.2 MHz
- Power handling shall be >= +29 dBm at 1 dB compression
- ESD tolerance >= 500 V (HBM)
- Die yield on qualification lot >= 94%

6. TEST RESULTS

Measurement Date: 2024-11-19
Devices Tested: 72
Insertion Loss (meas): 2.48 dB
Return Loss (meas): 22.3 dB
Rejection (meas): 30.8 dB
Isolation (meas): 31.7 dB
Q Factor (meas): 1148
Group Delay Variation: 7.8 ns
Power Handling: +33 dBm
Die Yield: 83.8%

Result Classification: CONDITIONAL PASS

7. OBSERVATIONS AND FINDINGS

- a) The Lattice Filter design demonstrated acceptable performance for Band 66 (AWS).
- b) Insertion loss met target specification.
- c) Out-of-band rejection exceeded the minimum requirement.
- d) Slight frequency drift observed at +85°C; TC layer thickness may need adjustment.
- e) Cross-talk between adjacent filter channels was below -55 dB.

8. CORRECTIVE ACTIONS

- 1. Review near-band rejection margin for worst-case temperature.
- 2. Re-run EM simulation with refined mesh for better accuracy.
- 3. Schedule follow-up measurement after design tweak.

9. SIGN-OFF

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